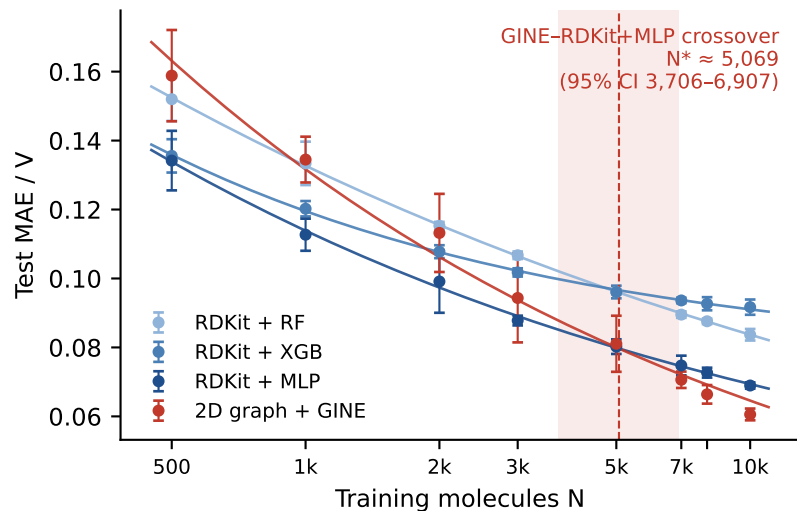
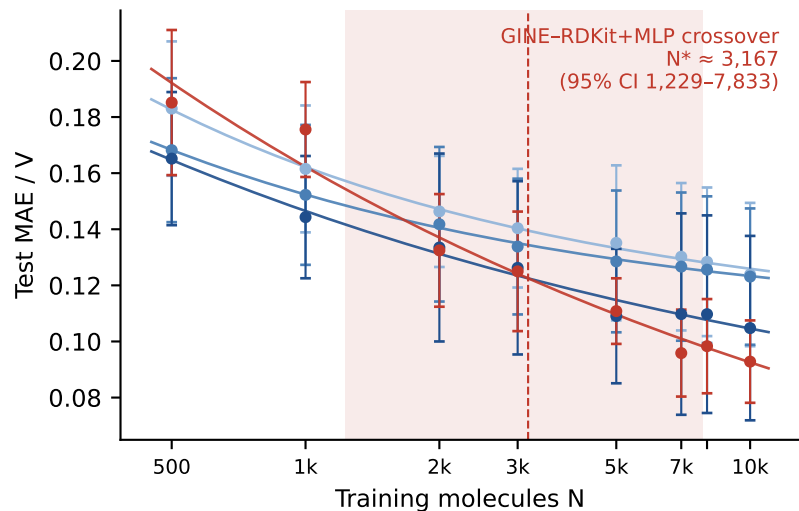


A Random split: power-law fits**B Scaffold split: power-law fits****C Random split: statistical top group**

RDKit + RF	3	3	4	4	4	3	3	3
RDKit + XGB	2	2	2	3	3	4	4	4
RDKit + MLP	1	1	1	1	1	2	2	2
Morgan + RF	6	5	5	5	5	5	5	5
Morgan + XGB	5	6	6	7	7	7	7	7
Morgan + MLP	7	7	7	6	6	6	6	6
2D graph + GINE	4	4	3	2	2	1	1	1
	500	1k	2k	3k	5k	7k	8k	10k

Training molecules N

D Scaffold split: statistical top group

RDKit + RF	3	3	4	4	4	4	4	4
RDKit + XGB	2	2	3	3	3	3	3	3
RDKit + MLP	1	1	2	2	1	2	2	2
Morgan + RF	6	6	5	5	5	6	6	6
Morgan + XGB	5	5	6	6	7	7	7	7
Morgan + MLP	7	7	7	7	6	5	5	5
2D graph + GINE	4	4	1	1	2	1	1	1
	500	1k	2k	3k	5k	7k	8k	10k

Training molecules N

C-D: numbers are mean-MAE ranks; dark outline = rank 1; shaded = not statistically separable from rank 1 (all five seeds agree in sign and paired t-test $p < 0.05$ required to separate); dark red = unique winner.